

Name:

Date:

Indigenous Environmental Data Science

Module 1: Monitoring Fisheries for Heavy Metal Contaminants

Task

☐ Check when complete, OR
think  and write  a response when prompted.

Lesson 1.1: R Basics and Calculating Screening Values

Learning Goals:

- Identify the function name that loads packages
- Recognize objects including variables, vectors, and data frames
- Calculate values using mathematical operations and functions
- Generate a scatterplot using ggplot2
- Describe the relationships between body weight, weekly servings, and screening values (safe limits)

Loading Packages

A toolbox (package) contains tools (functions) that perform specific math tasks that are typically used in data science.

 Question: What is the name of the function that loads a package of interest?

 Answer: _____

Creating Objects in R

An object is like a container or a box that holds data or information. You can give this box a name, and then use that name to look at or change what's inside the box later.

 Question: Which objects were created to calculate consumption rate?

 Answer: _____

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Calculating Screening Values for Mercury in Fish

Calculate the safe limit of consumption of fish per week for someone that is 160 lbs:

*Screening value = reference dose * weight * 0.45/(days per week)*

🖋 160 lbs = _____

🖋 Change the value for weight or days per week, and record the changes to the safe limit in the table below.

Weight (lbs)	Number of days per week	Screening Value (µg/g/day)

Creating Vectors

A vector is a list of the same type of object (i.e., numbers, words, TRUE/FALSE). A vector is an object, but not all objects are vectors.

🧠 Question: What function was used to create the vectors?

🖋 Answer: _____

Creating a Data Frame from Vectors

A data frame in R is a way to organize data in a table, similar to a spreadsheet. In a data frame,

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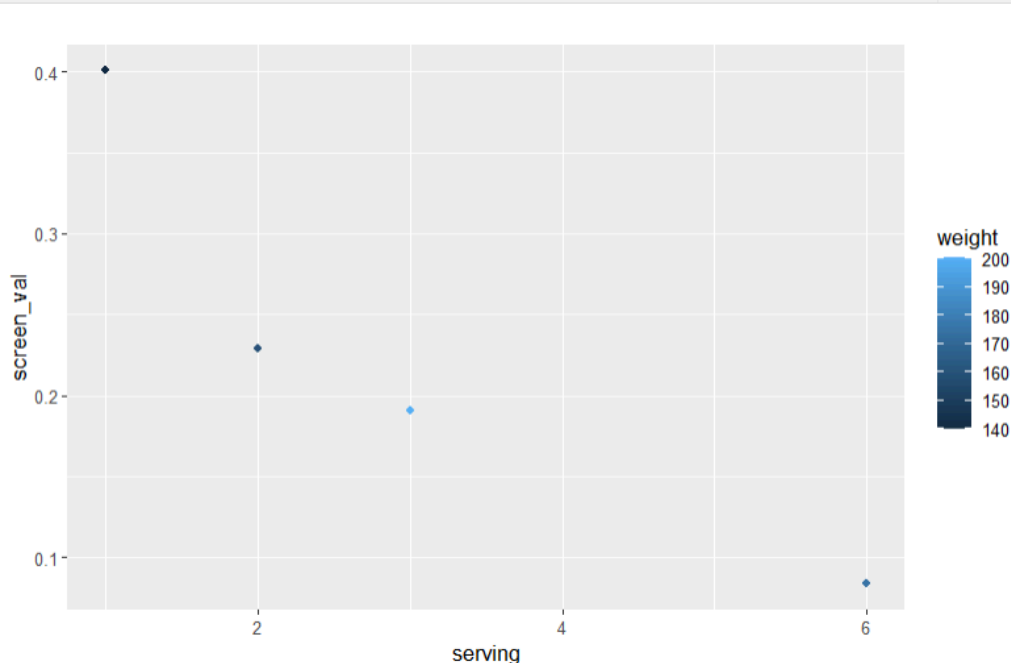
each column holds a group of related values (like weight, servings, and screening value), and each row represents one item or person.

🧠 Question: Provide an example of a time when you have created a data frame. It could be in any class and be made by hand or using something like Microsoft Excel or Google Sheets.

🖋 Answer: _____

Visualizing Screening Value vs. Servings per Week with ggplot2

```
178 # plot to see how servings per week and screening value are related (will practice more ggplot later)
179 ggplot(sv_df) + geom_point(aes(x=serving, y=screen_val, col=weight))
180
181
```



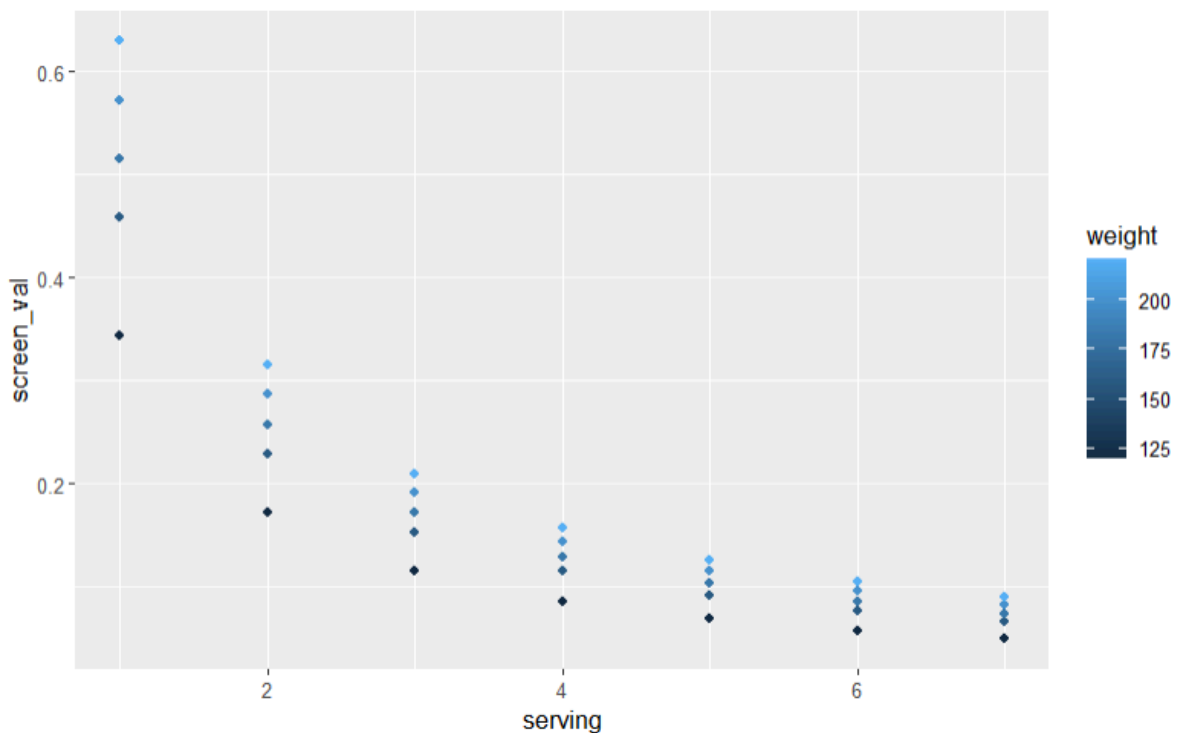
🧠 Question: What is the relationship between screening value and the number of servings?

🖋 Answer: _____

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```
187 sv_df_long = data.frame(weight, serving)
188 sv_df_long$screen_val = RfD*sv_df_long$weight*0.45/(110*sv_df_long$serving/7)
189 ggplot(sv_df_long) + geom_point(aes(x=serving, y=screen_val, col=weight))
190
191 ▲ ...
```



🧠 Question: How would you describe this graph and the relationship between screening value, consumption rate, and weight? What happens to the distribution in screening values as the servings per week increase?

🖋 Answer:

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🧠 Question: With this information in mind, consider how to make decisions with the data on mercury content in fish—how should we consider the screening value for different ages? For example, what screening value would be safe for children, who weigh less than the average adult?

🖋 Answer:

Lesson 1.2: Loading and Exploring 1998 Data

Learning Goals:

- Load in the 1998 fish data set
- View the entire data set in spreadsheet form
- View a truncated version of the data set
- Identify the number of rows and columns in the 1998 fish data set

▶ Run code to load 1998 fish data

☐

▶ Run code to view the 1998 fish data in spreadsheet form

☐

▶ Run code to view the first 6 rows of the 1998 fish data

☐

🧠 Question: What are the columns in the R dataframe called fish_1998?

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🖋 Answer:

🧠 Question: Identify at least 2 questions that the tribal fishery managers could answer with this data.

🖋 Answer:

🧠 Question: How many fish did the managers measure mercury on in 1998?

🖋 Answer: _____

Lesson 1.3: Calculating Descriptive Statistics

Learning Goals:

- Identify the descriptive statistics associated with the 1998 fish data set
- Obtain descriptive statistics for a vector
- Interpret the summary statistics for the 1998 fish data set

▶ Run code to determine the mean.

☐

🧠 Question: What is the mean level of mercury in the data?

🖋 Answer: _____

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 Question: Explain why knowing the mean (average) mercury level in fish could be useful for fishery managers and the community.

 Answer:

 Run code to determine the median.

☐

 Question: What is the median level of mercury in the data?

 Answer: _____

 Question: How would we interpret these median and mean values? Should the K'avi community be concerned?

 Answer:

 Run code to determine the minimum and maximum values.

☐

 Question: What is the minimum level of mercury in the data?

 Answer: _____

 Question: What is the maximum level of mercury in the data?

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🖋 Answer: _____

🧠 Question: How would we interpret these maximum and minimum values? Are they as important as the mean (average)?

🖋 Answer:

▶ Run code to determine the descriptive statistics, which includes the minimum, maximum, mean, and median, as well as the 1st and 3rd quartile.



🧠 Question: Write down your descriptive statistics results below.

🖋 Answer:

Min	1st Qu.	Median	Mean	3rd Qu.	Max

🧠 Question: We have already explored how unsafe levels of mercury depend on consumption rates, and that the FDA recommends avoiding anything above 0.46 $\mu\text{g/g/day}$. At what point in the distribution does the screening value exceed 0.46 $\mu\text{g/g/day}$?

🖋 Answer:

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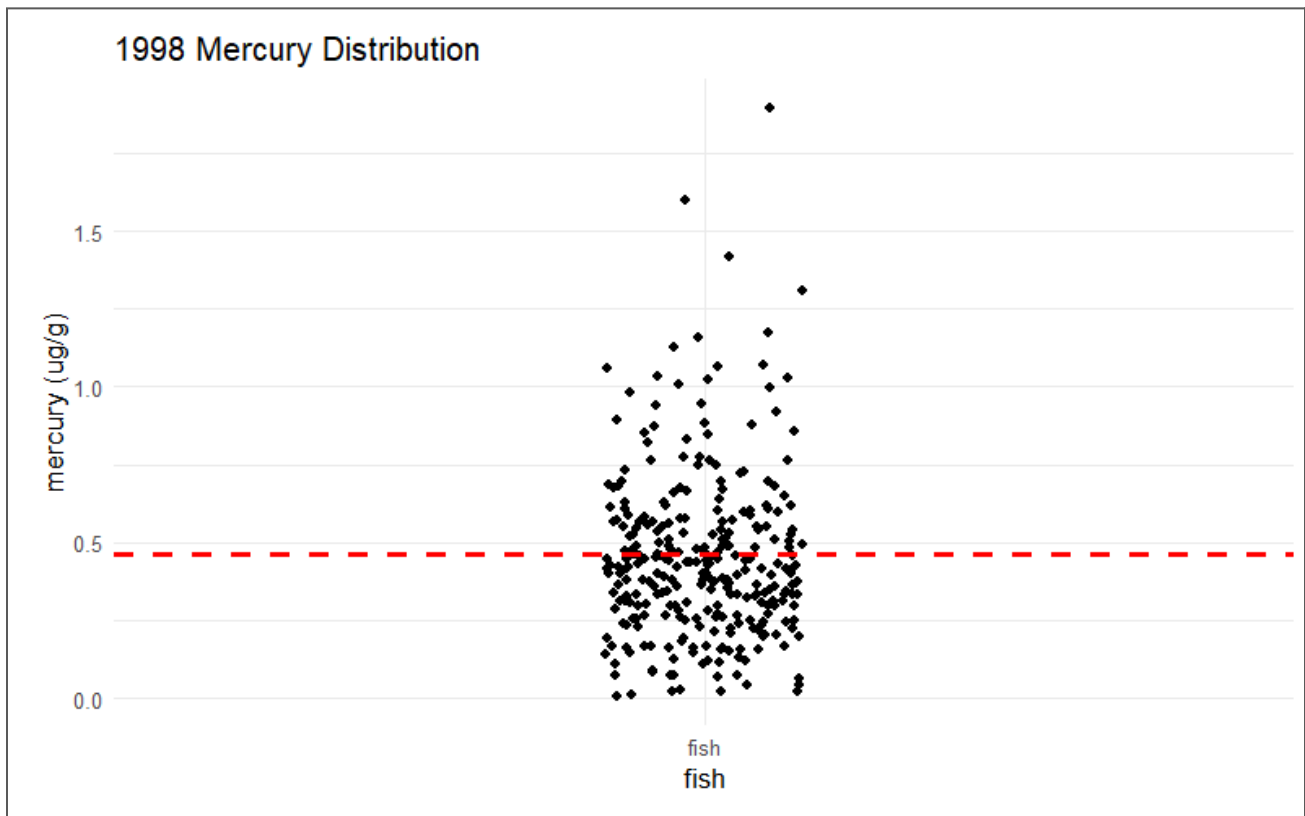
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Lesson 4: Plotting Mercury Concentrations with ggplot

Goals:

- Create a graph using the ggplot2 package to visualize mercury concentrations
- Analyze the distribution of mercury in fish to determine how often fish have mercury levels above the EPA recommended safe level in 1998

🧠 Question: What does this distribution tell us about the mercury levels in the fish?



🖋 Answer:

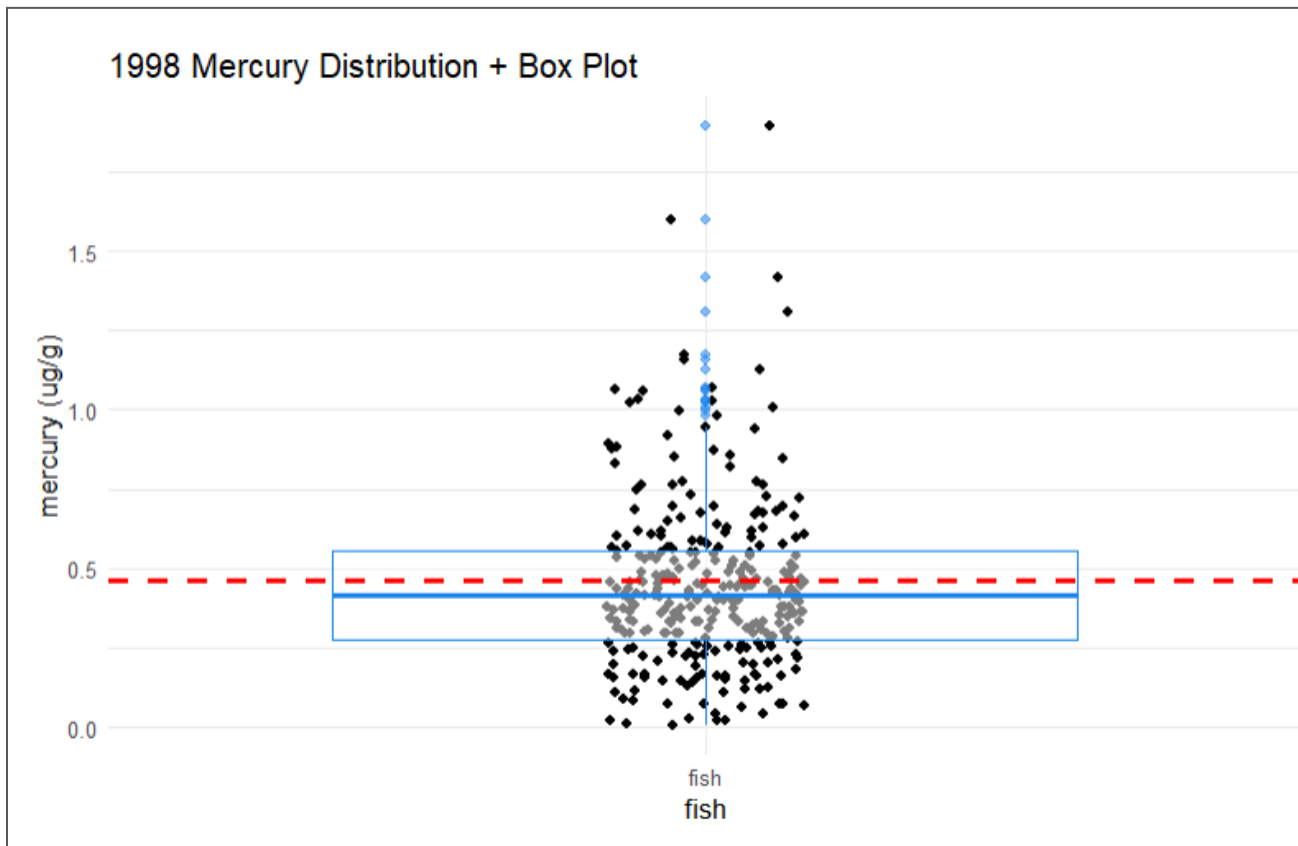
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🧠 Question: Look back at the min, max, median, and mean values that were calculated above. Where would these values fall on this graph? Are the mean and median values above or below the level suggested by the EPA?

🖋 Answer:

🧠 Question: From looking at this graph, estimate at what point in the distribution do the observed values exceed the screening value of 0.46? (i.e. which quartile, above or below mean)



🖋 Answer:

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🧠 Question: If so, is it in most fish, or only some? Did this graph tell you anything different than the summary statistics, or did it just tell you the same information in a different way?

🖋 Answer:

🧠 Question: In 1998, was it safe for members of the K'avi community to eat fish from these waters?

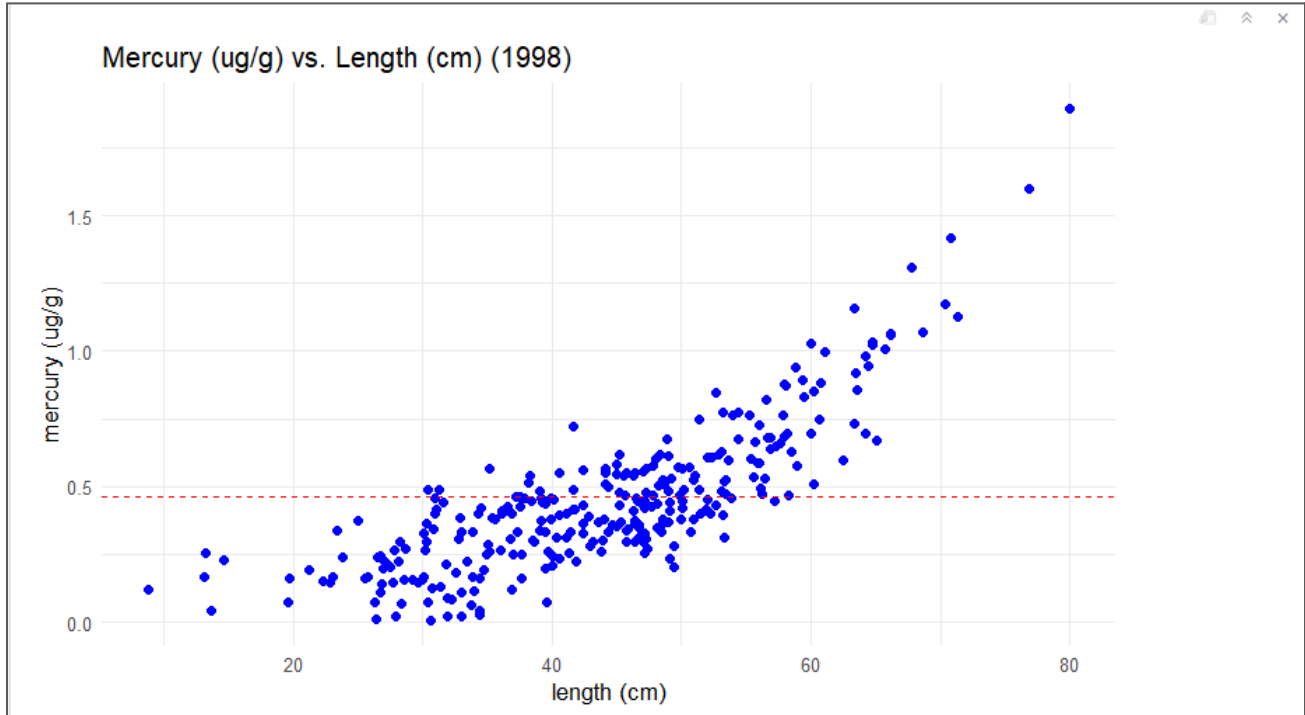
🖋 Answer:

🧠 Question: How comfortable would you have been in 1998 about eating a fish that you caught?

🖋 Answer:

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🧠 Question: What does this tell us about the relationship between fish length and mercury levels?

✍️ Answer:

🧠 Question: Are there certain lengths of fish that exceed the FDA screening value?

✍️ Answer:

🧠 Question: If you caught a fish in 1998, would this graph help you decide whether the fish was safe?

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🖋️ Answer:

Lesson 5: Putting It All Together to Explore the 2023 Fish Survey

Goals:

- Load in the 2023 fish data set
- View the entire 2023 fish data set in spreadsheet form
- View a truncated version of the 2023 fish data set
- Identify the number of rows and columns in the 2023 fish data set
- Identify the descriptive statistics associated with the 2023 fish data set
- Analyze data points from the 2023 fish data set to answer the following questions for tribal fishery managers:
 - Do the mercury levels in local fish belly fat exceed the maximum safe level of mercury to be consumed?
- Create a plot of the relationship between mercury and fish length in 1998
- Create a plot of the relationship between mercury and fish length in 2023
- Plot and analyze the 1998 and 2023 fish data side-by-side
 - Is there a relationship between the amount of mercury in fish and fish length?
- Interpret the results to describe the efficacy of removing contaminated sediment

Load “tidyverse” package

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Install and load “patchwork” package

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▶ Run code to load 2023 fish data

☐

Write code for head() prompt to see the first six rows of data

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🧠 Question: What are the variables in the 2023 fish data set?

🖋️ Answer:

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Write code for dim() to identify the number of rows and columns.

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🧠 Question: How many rows and columns are in the data set?

🖋 Answer:

🧠 Question: Write down your descriptive statistics results for 2023 below.

🖋 Answer:

Year	Min	1st Qu.	Median	Mean	3rd Qu.	Max
1998	0.007764	0.278103	0.413540	0.448118	0.557474	1.894938
2023						

🧠 Question: ??

🧠 Question: How do the mean and median values of mercury in 2023 compare to the EPA's standard?

🖋 Answer:

🧠 Question: Do the descriptive statistics suggest that the mercury levels in fish have improved

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since 1998?

🖋 Answer:

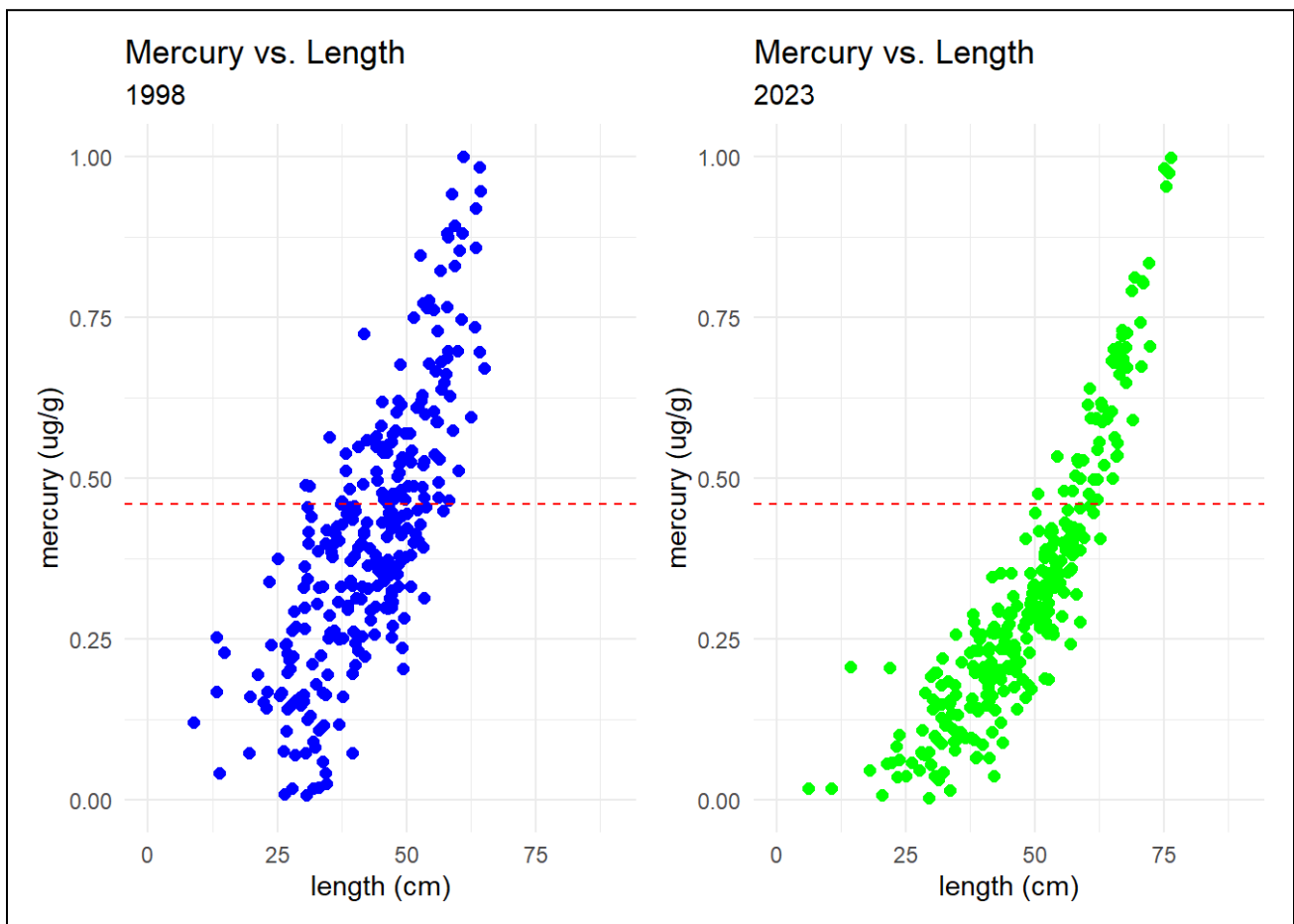
▶ Run code to plot 1998 fish data.

☐

▶ Run code to plot 2023 fish data.

☐

▶ Run code to plot 1998 and 2023 side-by-side. This is why we needed the “patchwork” package.

☐

🧠 Question: What is the safest large size fish to eat based on the 2023 dataset? How does this

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think 🧠 and write ✎ a response when prompted.

compare to the safest largest size fish that could be eaten based on the 1998 dataset?

✎ Answer:

🧠 Question: Based on these plots, do you think the efforts since 1998 to clean mercury from the sediments were successful?

✎ Answer:

🧠 Work with a group to describe the evidence from these graphs for the success of the clean-up effort in a way that would be useful for the K'avi community.

✎ Answer:
